

Practice Question Set

1. What is the basic Input → Processing → Output paradigm?
 2. What is the purpose of `stdio.h`?
 3. What does `printf()` stand for?
 4. What is the syntax of `printf()`?
 5. What is the purpose of `return 0;`?
 6. What is an escape sequence?
 7. What is the use of `\n`?
 8. What is a format specifier?
 9. What are the format specifiers for `int`, `char`, `float`, `double`, and `string`?
 10. What does `scanf()` stand for?
 11. What is the syntax of `scanf()`?
 12. Why is the address operator (`&`) used with `scanf()`?
 13. How can multiple values be accepted using `scanf()`?
 14. How can multiple values be displayed using `printf()`?
 15. Differentiate between `printf()` and `scanf()`.
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Important Interview/Viva Questions

16. Why is `stdio.h` required?
17. What is the difference between `%d` and `%i`?

18. Why is & necessary in scanf()?
 19. Why is & not used with printf()?
 20. What happens if & is omitted in scanf()?
 21. What is the difference between %f and %lf?
 22. Why is printf() called a formatted output function?
 23. Why is scanf() called a formatted input function?
 24. Explain multiple input and output with an example.
 25. What is the significance of return 0;?
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Answers :

1. What is the Input → Processing → Output paradigm?

Answer:

Every program follows the sequence:

Input → Processing → Output

- Input: Data entered by user.
 - Processing: Computation performed.
 - Output: Result displayed.
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2. What is the purpose of `stdio.h`?

Answer:

`stdio.h` stands for **Standard Input Output Header File**.

It provides predefined functions such as:

- `printf()`

- `scanf()`
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3. What does `printf()` stand for?

Answer:

`printf()` stands for **Print Formatted**.

It sends output to the monitor screen.

4. What is the syntax of `printf()`?

Answer:

```
printf("Message");
```

or

```
printf("Format Specifier", variable);
```

Example:

```
printf("Age = %d", age);
```

5. What is the purpose of `return 0;`?

Answer:

`return 0;` indicates that the program has executed successfully and terminates normally.

6. What is an escape sequence?

Answer:

Escape sequences are special characters preceded by a backslash (`\`) used for formatting output.

Example:

- `\n`

- \t

7. What is the use of \n?

Answer:

\n moves the cursor to the next line.

Example:

```
printf("Hello\n");  
printf("World");
```

Output:

```
Hello  
World
```

8. What is a format specifier?

Answer:

A format specifier is a placeholder beginning with % that tells printf() and scanf() what type of data is being processed.

9. What are the common format specifiers?

Data Type	Format Specifier
int	%d or %i
unsigned int	%u
long int	%ld
char	%c
float	%f
double	%lf
string	%s

10. What does scanf() stand for?

Answer:

scanf() stands for **Scan Formatted**.

It accepts input from the keyboard.

11. What is the syntax of scanf()?

Answer:

```
scanf("format_specifier", &variable);
```

Example:

```
scanf("%d", &age);
```

12. Why is the address operator (&) used with scanf()?

Answer:

Because scanf() needs the memory address of the variable to store the input value.

Example:

```
scanf("%d", &age);
```

&age gives the address of variable age.

13. How can multiple values be accepted using scanf()?

Answer:

By using multiple format specifiers.

Example:

```
scanf("%d%d", &a, &b);
```

14. How can multiple values be displayed using printf()?

Answer:

By using multiple format specifiers.

Example:

```
printf("%d %d", a, b);
```

15. Differentiate between `printf()` and `scanf()`.

<code>printf()</code>	<code>scanf()</code>
Output function	Input function
Print Formatted	Scan Formatted
Displays data	Accepts data
Does not require &	Requires &

Important Interview Answers

16. Why is `stdio.h` required?

Answer:

Because functions like `printf()` and `scanf()` are declared inside `stdio.h`.

Without it, the compiler will not recognize these functions.

17. What is the difference between `%d` and `%i`?

Answer:

For integers, `%d` and `%i` behave almost the same and are used to represent signed integers.

Example:

```
printf("%d", x);  
printf("%i", x);
```

Both produce the same output.

18. Why is & necessary in `scanf()`?

Answer:

Because `scanf()` needs the address of the variable where the input value should be stored.

19. Why is & not used with `printf()`?

Answer:

Because `printf()` only reads the value of the variable and displays it; it does not modify the variable.

Example:

```
printf("%d", age);
```

20. What happens if & is omitted in `scanf()`?

Answer:

The compiler may generate an error or the program may exhibit undefined behavior because `scanf()` will not know where to store the input.

Incorrect:

```
scanf("%d", age);
```

Correct:

```
scanf("%d", &age);
```

21. What is the difference between `%f` and `%lf`?

`%f`

`%lf`

Used for float

Used for double

Single precision Double precision

22. Why is `printf()` called a formatted output function?

Answer:

Because it displays data according to format specifiers.

Example:

```
printf("Age = %d", age);
```

23. Why is `scanf()` called a formatted input function?

Answer:

Because it accepts input according to format specifiers.

Example:

```
scanf("%f", &marks);
```

24. Explain multiple input and output with an example.

Answer:

```
int a, b;
```

```
scanf("%d%d", &a, &b);
```

```
printf("%d %d", a, b);
```

If user enters:

```
10 20
```

Output:

```
10 20
```

25. What is the significance of `return 0;`?

Answer:

It informs the operating system that the program terminated successfully without errors.